

# Adverbial modification and anti-roqativity

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Why is (1) ungrammatical?

- (1) \*Ayesha believes who won the match. (cf. Zuber 1982)

One influential approach (Mayr 2019, sec. 4.3.2; Theiler et al. 2019, sec. 3.1.5; Roberts 2019, 674) explains its ungrammaticality with two assumptions:

- (i) *Believe* has an excluded middle presupposition, which also explains why *believe* is neg-raising, to the effect that *believe*'s external argument is opinionated about the question its embedded clause denotes;
- (ii) If an utterance contains a clausal constituent with (a logical skeleton with) presupposed content  $p$  and at-issue content  $a$ , it's ungrammatical if  $p \subseteq a$  (the presupposition entails the at-issue content) or  $p \cap a = \emptyset$  (the presupposition is inconsistent with the at-issue content).

These assumptions are then used as follows. By (i), (1)'s presupposed content is that Ayesha is opinionated about the question of who won the match. But, that's also (1)'s at-issue content. So, its presupposed content is a subset of its at-issue content. So, by (ii), (1) is ungrammatical.

Manner adverbs raise a challenge for this approach. Take:

- (2) \*Bastian mistakenly believes who won the match.

where *mistakenly* gets a manner reading (rather than a clausal one) and entails that Bastian believes in a mistaken way (not that it's a mistake for Bastian to believe). (Kubota 2015 1024-5 discusses manner and clausal readings.) (2) is ungrammatical, just like (1). Yet (2)'s presupposed content isn't a subset of its at-issue content, nor is their intersection empty. (2)'s presupposed content is that Bastian is opinionated; its at-issue content is that he's opinionated in a mistaken way.

Now, if *Bastian believes who won the match* were a clausal constituent of (2), we could explain (2)'s ungrammaticality. But on *mistakenly*'s manner reading, *mistakenly* modifies the VP *believe who won the match* before we get a clause. So, (2)'s only clausal constituent, besides (2) itself, is *who won the match*. But that constituent's presupposed content (that someone won the match) isn't a subset of its at-issue content (a set of propositions); and even if the intersection of these contents is empty, that can't explain (2)'s ungrammaticality because it would make any utterance containing *who won the match* ungrammatical.

One reply to my challenge says (2) is ungrammatical because the VP *believe who won the match* is (cf. Champollion 2015), or is extendable into, a clausal constituent whose presupposed content is a subset of its at-issue content. However, this predicts the utterances in (3) to be ungrammatical, although the first is generally felicitous and the second only odd if contextually unsupported (Roberts 2019, 676).

- (3) A: I can't believe who won the election.  
 B:  $[I]_F$  can believe who won the election.

Another reply says (2) is ungrammatical because *mistakenly*'s manner reading modifies *believe*'s presupposition so that *mistakenly believe* presupposes that its external argument is mistakenly opinionated about the question its embedded clause denotes. Given this, (2)'s presupposed content is a subset of its at-issue content. However, since *believe*'s excluded middle presupposition explains why it's neg-raising, this reply requires making *mistakenly believe* neg-raising too. Yet (4) doesn't trigger an inference that Bastian mistakenly believes that Ayesha didn't win the match (Heycock 2005, 371).

- (4) Bastian doesn't mistakenly believe that Ayesha won the match.

Clearly, the replies just discussed aren't exhaustive. Still, their problems illustrate why (2)'s ungrammaticality is a significant datum for understanding *believe*'s anti-rogativity.

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